UEC859: INTEGRATED SYSTEM DESIGN

L	T	P	Cr
2	0	2	3.0

Course Objective: To enhance comprehension capabilities of students through understanding on the use of **HDL (Verilog and VHDL)** for the design, synthesis, modeling, and testing of VLSI devices. These are IEEE standards that are used by engineers to efficiently design and analyze complex digital designs.

Basic Digital Circuits: Lexical Elements and data types, program skeleton, structural, dataflow and behavioural descriptions, testbench.
RTL Combinational circuit: Operators, Block statement, Concurrent assignment statements, Modelling with a process, Routing circuit with if and case statements, Constants and Generics
Regular Sequential Circuit: HDL code of Flip flops and Registers, simple design examples, testbench for sequential circuits, case study
FSM: Mealy and Moore FSMs, Design Examples
Synthesis: Register Transfer level description, Timing and CLO sck Constraints, technology libraries, Translation, Boolean optimization, Factoring, Mapping to gates
Xilinx FPGA Implementation Memory: Method to incorporate memory modules, HDL templates for memory interface

Familiarization with standards: *IEEE 1164-1993, IEEE 1076-2019.*

Laboratory Work: Modeling and simulation of all Verilog and VHDL constructs using, their testing by modeling and simulating test benches, Logic Synthesis using FPGA Advantage, Mapping on FPGA Boards.
Micro Poject: Design & Simulate a digital system in VHDL or Verilog and its implementation FPGA board.

Course Learning Outcomes (CLOs): The student will be able to:
1. Build a synchronous system in hdl and verify its performance.
2. Build and test complex FSMs
3. Automate testbenches for automatic pass/fail
4. Make design decisions for fixed point implementations given constraints
5. Analyse memory usage/requirements for FPGA
6. Target sequential designs to FPGA

Text Books:

1. *Bhaskar, J., A VHDL Primer, Pearson Education/ Prentice Hall (2006)3rd Ed.*
2. *Palnitkar, Samir, Verilog HDL, Prentice Hall, 2nd Edition,*

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

Reference Books:

1. *Ashenden, P., The Designer's Guide To VHDL, Elsevier (2008) 3rd Ed.*
2. *Donald E. Thomas, Philip R. Moorby, Donald B. Thomas, The Verilog HDL, Kluwer Academic Publication, 5th Edition, 2002,*
3. *Chu Pong P., FPGA Prototyping by VHDL / Verilog Examples, Wiley (2008)*

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1	MST+EST	70
2	Sessional (May include Assignments/Projects/Tutorials/ Quizzes/Lab Evaluations)	30